Printed Pages : 4

Semester : V

TCS05027

Roll No. 2108L22037

Computer Science and Engineering

December-2023

DESIGN & ANALYSIS OF ALGORITHM

Maximum Marks : 70

Time Allowed : 3 Hours

Note: i) Attempt all the questions

ii) Maximum marks assigned to each question is indicated.

Section - A

Q.1. Attempt all questions (2 marks questions). **(2x5=10)**

(a) Calcualte time complexity for the given pseudocode. (BT-4, CO-1, PO-1)

```
for (i=1; i<n; i=i*2)
{statements;
}
```

(b) Explain min-heap ordering property. (BT-1, CO-2, PO-1)

(c) State the principle of optimality. (BT-1, CO-3, PO-1)

(d) Define chromatic number(BT-2, CO-4, PO-1)

(e) Discuss time complexity of naive string match algorithm. (BT-1, CO-5, PO-2)

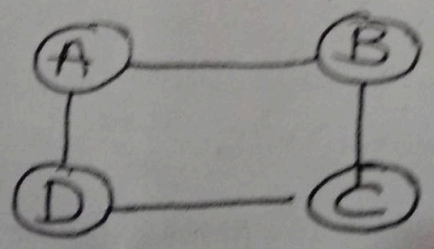
Attempt all questions (short answer questions). (4x5=20)

(f) Write a brief description on Asymptotic notation. (BT-2, CO-1, PO-1)

(g) State the properties of a Red Black tree and also calculate its time complexity. (BT-3, CO-2, PO-1)

(h) Differentiate between dynamic programming and greedy approach. (BT-2, CO-3, PO-2)

(i) Explain Graph coloring Problem. Determine the number of colors used for the graph given below. (BT-1, CO-4, PO-1)



(j) Explain Knuth-Morris Pratt algorithm. Draw π table for the pattern ababd. (BT-3, CO-5, PO-3)

(c) State the principle of optimality. (BT-1, CO-3, PO-1)

(d) Define chromatic number(BT-2, CO-4, PO-1)

(e) Discuss time complexity of naive string match algorithm. (BT-1, CO-5, PO-2)

Attempt all questions (short answer questions). (4x5=20)

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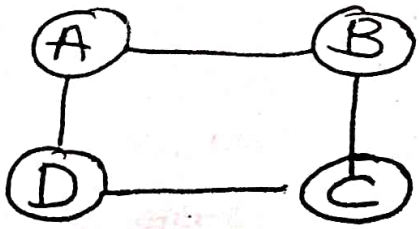


(j) Explain Knuth-Morris Pratt algorithm. Draw π table for the pattern ababd. (BT-3, CO-5, PO-3)

Section - B

Q.2. Calculate the time complexity for the given recurrence relation using substitution method. 8

$T(n) = T(n-1) + \log n$ (BT-4, CO-1, PO-3)

OR

Explain merge sort algorithm and trace this algorithm for n = 8 elements. (BT-2, CO-1, PO-1)

{24, 12, 35, 23, 45, 34, 20, 48}

Q.3 Let A {7, 2, 4, 17, 1, 11, 6, 8, 15, 10, 20}. Draw binomial heap for above elements.8 (BT-1, CO-2, PO-2)

OR

Explain B-Tree and construct a B-Tree for following elements. 8(BT-1, CO-3, PO-2)

{5, 3, 21, 9, 1, 13, 2, 7, 10, 12, 4, 8}

Q.4. Find an optimal sequence to the n = 5 jobs where profits $(P_1, P_2, P_3, P_4, P_5) = (20, 15, 10, 5, 1)$ and deadlines (BT-2, CO-3, PO-3)

$(d_1, d_2, d_3, d_4, d_5) = (2, 2, 1, 3, 3)$

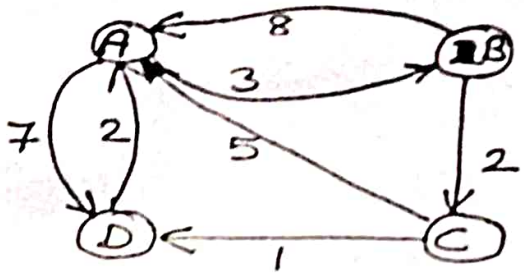
OR

Explain Longest common subsequence. Find the longest common subsequence for X = <A, B, C, B, D, A, B> and Y =<B, D, C, A, B, A>

(BT-2, CO-3, PO-3)

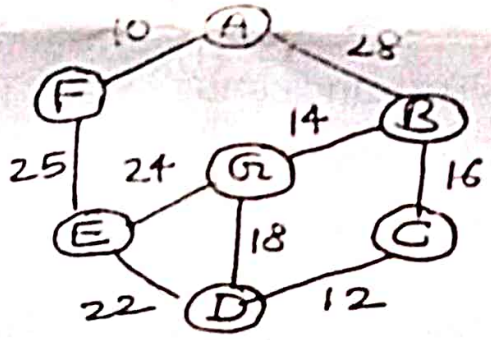
Q.5. Explain Floyd-Warshall Algorithm. Determine the shortest path for the given graph. 8

(BT-4, CO-4, PO-2)



OR

Explain a spanning tree. find the minimum cost of the given graph using Prim's algorithm. (BT-4, CO-4, PO-3)



Q.6. Explain Rabin Karp string matching algorithm. Also determine for q=11, how many valid and spurious hit found for given text and pattern P=26.

(BT-2, CO-5, PO-1)

T = 3 1 4 1 5 9 2 6 5 3 5 8 9 7 9 3

OR

Explain NPhard and NP complete in detail.

(BT-1, CO-5, PO-2)